Indian Space Olympiad (ISO) 2026

SYLLABUS - SENIOR LEVEL (CLASSES IX–X)

01. Fundamentals of Astronomy

Celestial coordinate systems, astronomical distances, telescopes, electromagnetic spectrum, observational astronomy, astronomical calculations (basic).

02. The Solar System

Formation and evolution of the Solar System, planetary geology, planetary atmospheres, Kepler's Laws, orbital motion, small Solar System bodies, exoplanets (basic).

03. Stars, Galaxies & the Universe

Stellar evolution, Hertzsprung-Russell Diagram (basic), white dwarfs, neutron stars, black holes, galaxies, Big Bang Theory, expanding Universe.

04. Earth & Moon

Earth system, atmosphere, magnetosphere, climate, Moon geology, Earth observation, space weather (basic).

05. Satellites & Space Technology

Satellite systems, communication, navigation, remote sensing, spacecraft subsystems, satellite applications.

06. Rockets & Launch Vehicles

Rocket propulsion, multistage launch vehicles, escape velocity, orbital insertion, cryogenic engines (concept), reusable launch vehicles.

07. ISRO & India's Space Missions

Launch vehicles, Chandrayaan programme, Mars Orbiter Mission, Aditya-L1, Gaganyaan, XPoSat, future missions and India's space roadmap.

08. Space Exploration

Planetary exploration, deep-space missions, human spaceflight, space stations, asteroid missions, future exploration.

09. Basic Physics for Space Science

Mechanics, gravitation, circular motion, waves, optics, electricity, magnetism, thermal physics, basic modern physics.

10. Scientific Reasoning & Logical Aptitude

Scientific reasoning, experimental interpretation, data analysis, mathematical reasoning, logical aptitude, application-based problem solving.

11. General Space Awareness

Current developments in astronomy, global space missions, commercial spaceflight, space policy, emerging technologies.